Smart Attendance Tracker with AI Risk Detection

Abstract

The Smart Attendance Tracker with AI Risk Detection is a Python-based application that automates student attendance management using CSV data storage. It records attendance, generates reports, calculates attendance percentages, and identifies students at risk of poor attendance using rule-based AI analysis.

Introduction

The project replaces manual attendance registers with an automated system that improves speed, accuracy, and record management.

Problem Statement

Manual attendance systems are time-consuming, prone to human error, difficult to analyse, and make identifying at-risk students challenging.

Objectives

Automate attendance recording.
Reduce human error.
Generate attendance reports.
Track attendance percentages.
Identify students at risk using AI logic.

Scope

The system supports student registration, attendance recording, report generation, attendance analysis, student search, and AI-based risk detection.

Technologies Used

Programming Language: Python.
Libraries: pandas, datetime, os.
Storage: CSV files.

Project Modules

main.py – Main application.
attendance.py – Attendance management.
reports.py – Report generation.
ai.py – AI risk detection.
utils.py – Helper functions.

Key Features

Register students.
Record attendance.
Prevent duplicate attendance.
Automatically use the current date.
Search students.
Generate attendance summaries.
Calculate attendance percentages.
AI risk detection.

Pain Points Addressed

Eliminates paper attendance registers.
Reduces recording errors.
Saves lecturers' time.
Improves record organisation.
Provides early warning for students with poor attendance.
Simplifies report generation.

Testing

The application was tested using sample student records to verify attendance recording, report generation, duplicate prevention, attendance calculations, and AI risk detection.

Conclusion

The Smart Attendance Tracker demonstrates how Python can automate attendance management while providing intelligent analysis to identify students requiring early intervention.